

Necessary Roads

Time limit: 1.00 s

Memory limit: 512 MB

There are n cities and m roads between them. There is a route between any two cities.

A road is called *necessary* if there is no route between some two cities after removing that road. Your task is to find all necessary roads.

Input

The first input line has two integers n and m : the number of cities and roads. The cities are numbered $1, 2, \dots, n$.

After this, there are m lines that describe the roads. Each line has two integers a and b : there is a road between cities a and b . There is at most one road between two cities, and every road connects two distinct cities.

Output

First print an integer k : the number of necessary roads. After that, print k lines that describe the roads. You may print the roads in any order.

Constraints

- $2 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input	Output
5 5 1 2 1 4 2 4 3 5 4 5	2 3 5 4 5